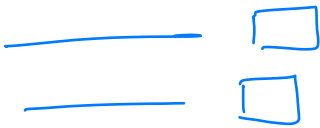


Agenda

RNN

{
Only I love you ★
I only love you ★
I love only you ★
I love you only ✕

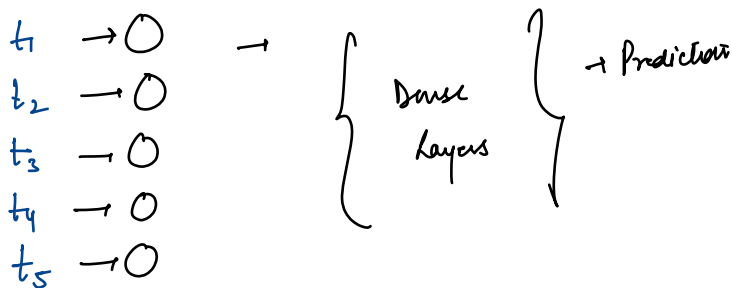
Deep Learning Engineers at BBC



Many to one
inputs output

one to Many
Image Caption

Traditional methods



① No temporal Component

② Padding

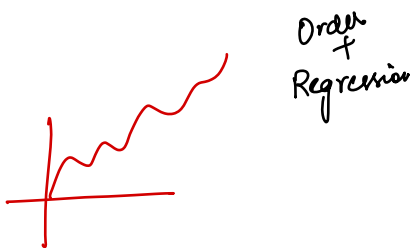
So words
title $\rightarrow$ 5 words

③ All words are isolated

motivation to have
Recurrent Neural Networks
 Word by word,
 maintains state,
 Uses shared weights

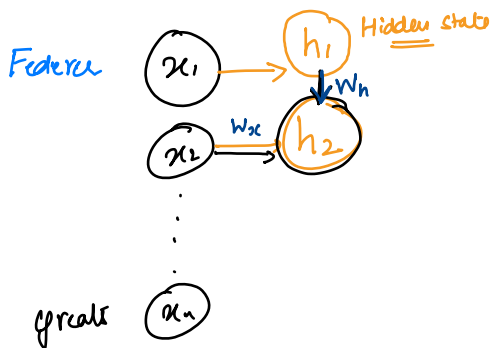
$\Rightarrow$ RNNs

Sequence



I only love you $\rightarrow$ classifier

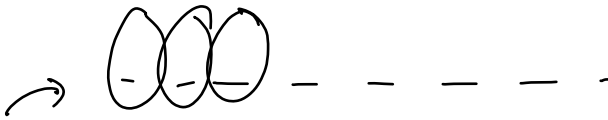
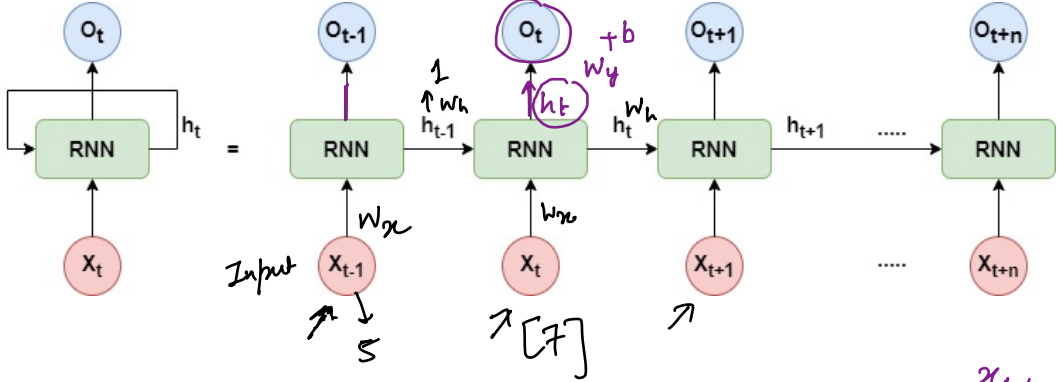
Federer joins all-time greats



Classical RNN cell

$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b_h)$$

$$O_t = \sigma(h_t w_y + b_y)$$



$$x_t = [1 \ 0]$$

$$h_{t-1} = [0.2 \ -0.1]$$

2D embedding

$$W_x = \begin{bmatrix} 0.5 & -0.3 \\ 0.8 & 0.1 \end{bmatrix}$$

$$W_h = \begin{bmatrix} 0.4 & 0.2 \\ -0.5 & 0.3 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b_h)$$

weight sharing

$$h_t = [0.51 \ 0.58]$$

$$h_{t-1} = 0.2 \times 1 + 0.3 \times 1 = 0.5$$

$$h_t = 0.2 \times 7 + 0.3 \times 0.5 = 1.475$$

Current input

Same weights are used at each time stamp.

→ One to One

→ One to Many

→ Many to One

→ Many to Many] language translation

$$\checkmark w_x \quad \checkmark w_h \quad \checkmark w_y$$

$$w_x = w_x - \alpha \frac{\partial L}{\partial w_x}$$

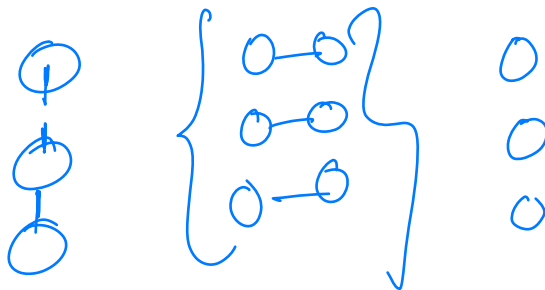
$$w_h = w_h - \alpha \frac{\partial L}{\partial w_h}$$

$$w_y = w_y - \alpha \frac{\partial L}{\partial w_y}$$

BPTT

(Back propagation through time)

↳ same weight across time stamp



RNN

↳ Bidirectional RNN

✓ Sachin Tendulkar was _____

✓ Sachin Tendulkar was _____ of Rajya Sabha

